

main.py > ...

```
1  import pandas as pd
2  import numpy as np
3  import matplotlib.pyplot as plt
4  import seaborn as sns
5
6  data = {
7      'Student_ID': [101,102,103,104,105,106,107,108,109,110],
8      'Name': ['Alice','Bob','Charlie','David','Eva',
9              'Frank','Grace','Helen','Ian','Jack'],
10     'Math': [85,78,92,67,88,73,95,81,69,90],
11     'Science': [90,74,89,70,85,76,98,80,65,93],
12     'English': [88,82,91,72,84,75,97,79,68,89]
13 }
14
15 df = pd.DataFrame(data)
16
17 print("Student Dataset")
18 print(df)
19
20 df['Total'] = df[['Math','Science','English']].sum(axis=1)
21 df['Average'] = round(df['Total']/3,2)
22
23 def assign_grade(avg):
24     if avg >= 90:
25         return 'A'
26     elif avg >= 80:
27         return 'B'
28     elif avg >= 70:
29         return 'C'
30     else:
31         return 'D'
32
33 df['Grade'] = df['Average'].apply(assign_grade)
34
35 print("\nStudent Performance Analysis")
36 print(df)
37
```

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```
38 print("\nStatistical Summary")
39 print(df[['Math', 'Science', 'English', 'Average']].describe())
40
41 top_student = df.loc[df['Average'].idxmax()]
42
43 print("\nTop Performer")
44 print(top_student)
45
46 subject_avg = df[['Math', 'Science', 'English']].mean()
47
48 print("\nSubject Wise Average")
49 print(subject_avg)
50
51 print("\nGrade Distribution")
52 print(df['Grade'].value_counts())
53
54 plt.figure(figsize=(8,5))
55 sns.barplot(x=subject_avg.index,
56            y=subject_avg.values)
57
58 plt.title("Average Score by Subject")
59 plt.ylabel("Average Marks")
60 plt.show()
61
62 plt.figure(figsize=(8,5))
63 sns.histplot(df['Average'],
64             bins=5,
65             kde=True)
66
67 plt.title("Distribution of Student Averages")
68 plt.xlabel("Average Marks")
69 plt.show()
70
71 plt.figure(figsize=(6,4))
72 sns.countplot(x='Grade',
73             data=df,
74             order=['A', 'B', 'C', 'D'])
```

```
71 plt.figure(figsize=(6,4))
72 sns.countplot(x='Grade',
73 |             |             | data=df,
74 |             |             | order=['A', 'B', 'C', 'D'])
75
76 plt.title("Grade Distribution")
77 plt.show()
78
79 df.to_csv("student_performance_report.csv",
80 |         |         | index=False)
81
82 print("\nReport saved as student_performance_report.csv")
```


Student Dataset

	Student_ID	Name	Math	Science	English
0	101	Alice	85	90	88
1	102	Bob	78	74	82
2	103	Charlie	92	89	91
3	104	David	67	70	72
4	105	Eva	88	85	84
5	106	Frank	73	76	75
6	107	Grace	95	98	97
7	108	Helen	81	80	79
8	109	Ian	69	65	68
9	110	Jack	90	93	89

Student Performance Analysis

	Student_ID	Name	Math	Science	English	Total	Average	Grade
0	101	Alice	85	90	88	263	87.67	B
1	102	Bob	78	74	82	234	78.00	C
2	103	Charlie	92	89	91	272	90.67	A
3	104	David	67	70	72	209	69.67	D
4	105	Eva	88	85	84	257	85.67	B
5	106	Frank	73	76	75	224	74.67	C
6	107	Grace	95	98	97	290	96.67	A
7	108	Helen	81	80	79	240	80.00	B
8	109	Ian	69	65	68	202	67.33	D
9	110	Jack	90	93	89	272	90.67	A

Statistical Summary

	Math	Science	English	Average
count	10.00000	10.000000	10.000000	10.000000
mean	81.80000	82.000000	82.500000	82.102000
std	9.82966	10.728985	9.107384	9.729835
min	67.00000	65.000000	68.000000	67.330000
25%	74.25000	74.500000	76.000000	75.502500
50%	83.00000	82.500000	83.000000	82.835000
75%	89.50000	89.750000	88.750000	89.920000
max	95.00000	98.000000	97.000000	96.670000

Top Performer

Student_ID	107
Name	Grace
Math	95
Science	98
English	97
Total	290
Average	96.67
Grade	A
Name: 6, dtype: object	

Subject Wise Average

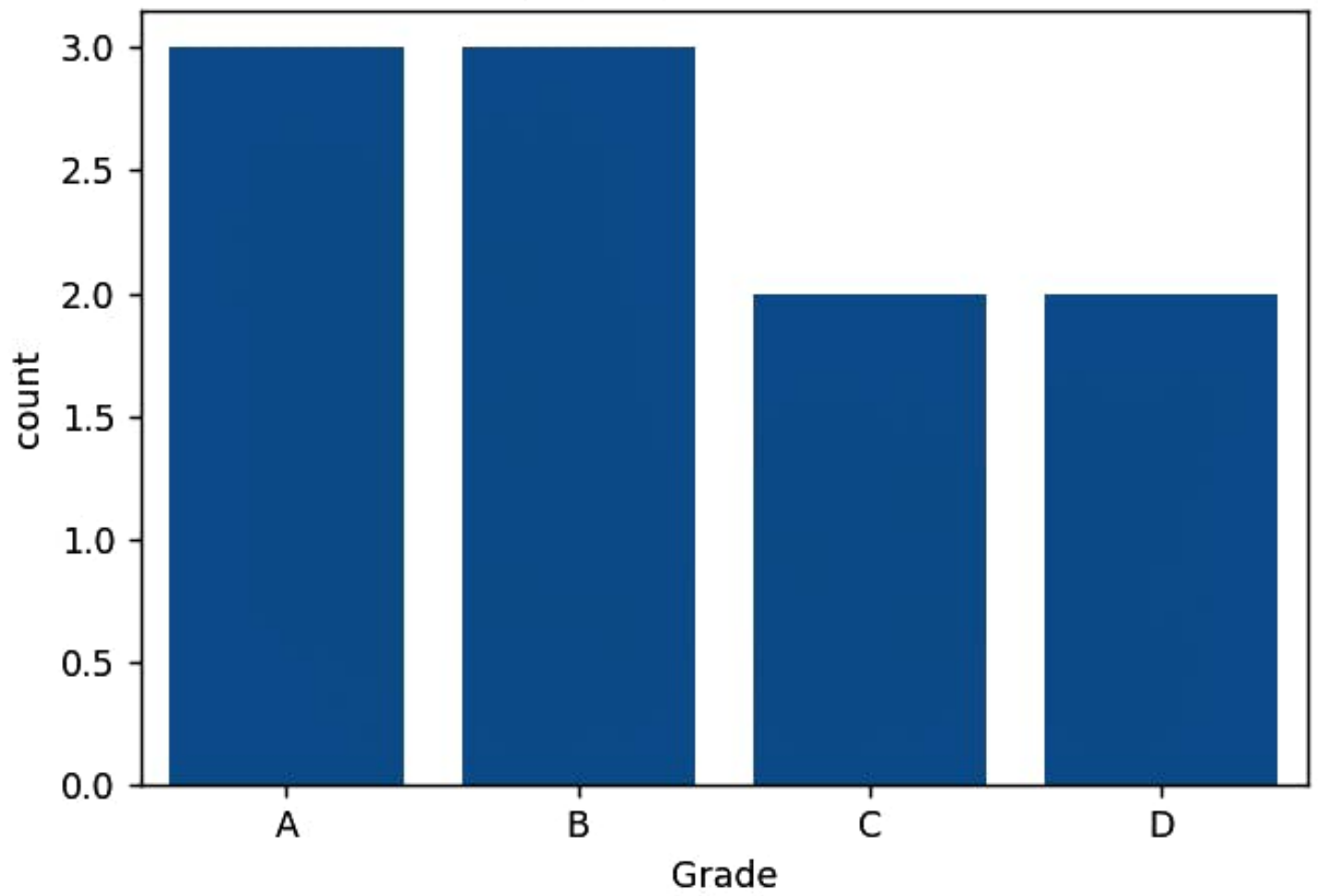
Math	81.8
Science	82.0
English	82.5
dtype: float64	

Grade Distribution

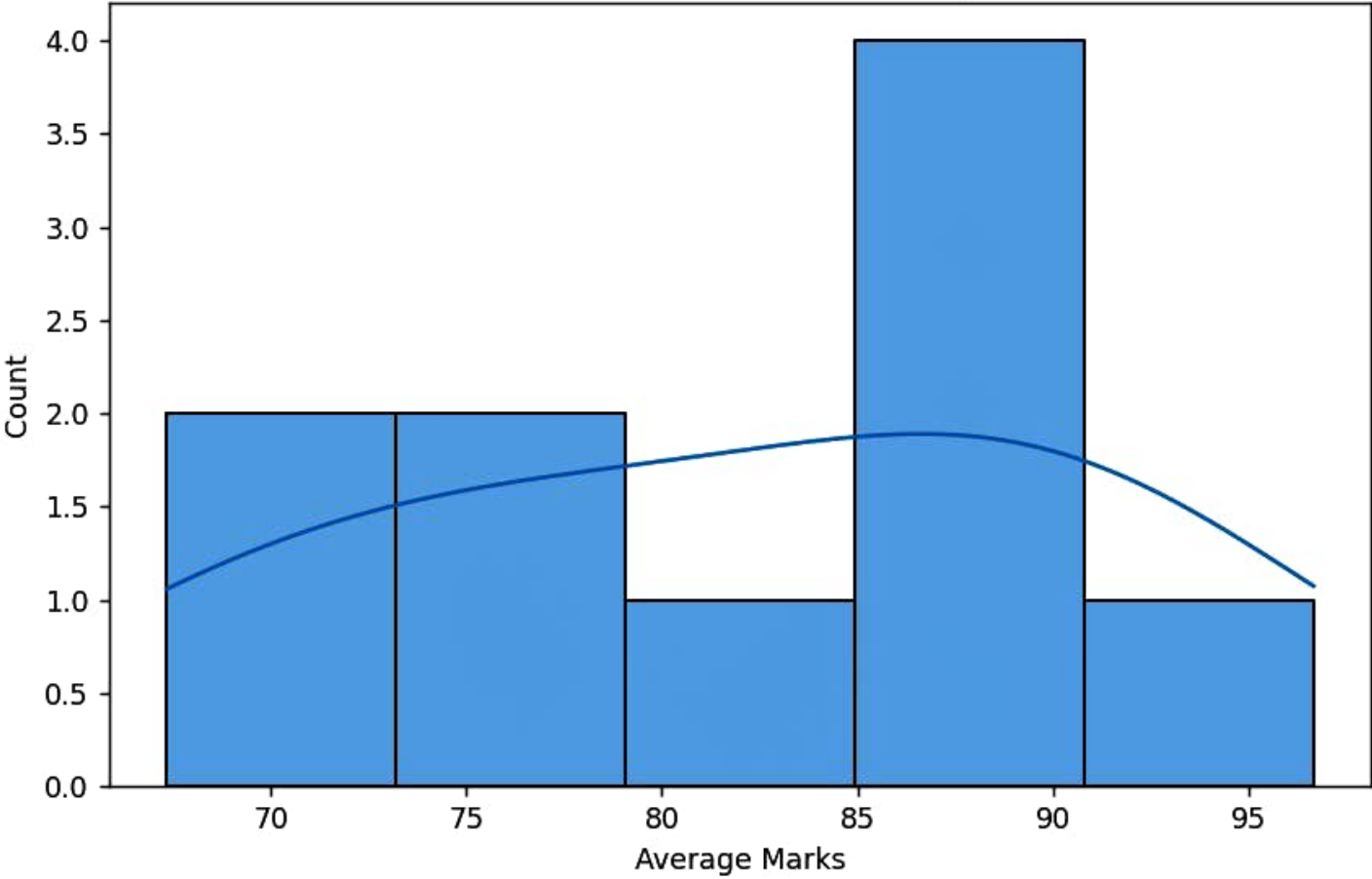
Grade	
B	3
A	3
C	2
D	2
Name: count, dtype: int64	

Report saved as student_performance_report.csv

Grade Distribution



Distribution of Student Averages



Average Score by Subject

